ECO 310: Problem 2b Graphical Analysis
WARP Violation Danger Zone

Visual Explanation

February 28, 2026

1 Problem Setup

Problem 2b

An elderly man previously consumed $(4, 2)$ when prices were $(1, 3)$. Prices are now $(3, 5)$. Which bundles would violate WARP (get him committed)?

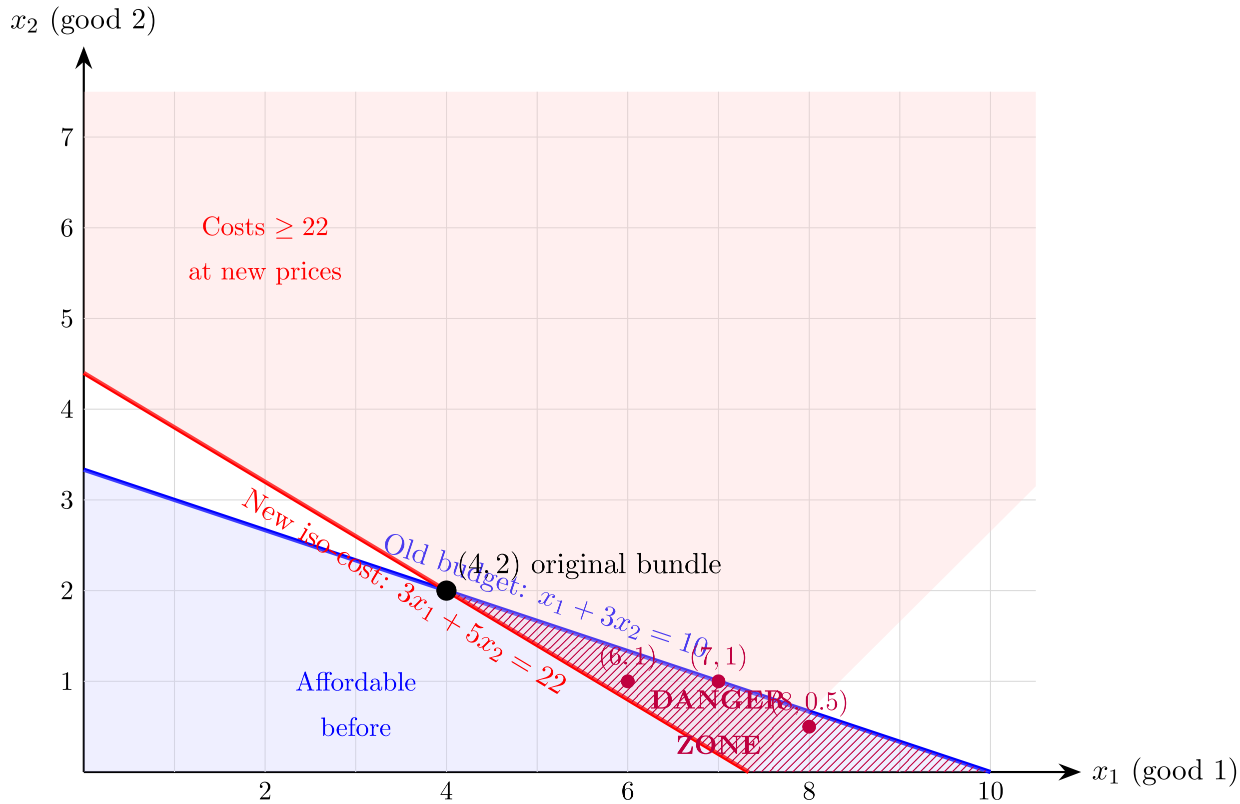
2 The Two Key Constraints

For a bundle (x_1, x_2) to create a WARP violation, it must satisfy BOTH:

1. **Inequality 1:** $x_1 + 3x_2 \leq 10$
   - Meaning: The bundle was **affordable at old prices**
   - Old cost: $1 \cdot x_1 + 3 \cdot x_2 \leq 10$ (his original budget)
   - He could have bought this bundle before, but chose $(4, 2)$ instead
   - This reveals: $(4, 2) \succ (x_1, x_2)$
2. **Inequality 2:** $3x_1 + 5x_2 \geq 22$
   - Meaning: The bundle costs **at least as much as** $(4, 2)$ **at new prices**
   - New cost of old bundle: $3(4) + 5(2) = 22$
   - If he buys (x_1, x_2) costing ≥ 22, then $(4, 2)$ was affordable
   - This reveals: $(x_1, x_2) \succ (4, 2)$

The Contradiction: Can't have $(4, 2) \succ (x_1, x_2)$ AND $(x_1, x_2) \succ (4, 2)$ simultaneously!

3 Visual Representation



4 Interpretation

4.1 The Blue Region (Affordable Before)

Everything below or on the blue line satisfies $x_1 + 3x_2 \leq 10$.

What this means: All these bundles were affordable when prices were $(1, 3)$ and his budget was 10.

Since he chose $(4, 2)$, he **revealed preference** that $(4, 2)$ is better than any other bundle in this region.

4.2 The Red Region (Expensive Now)

Everything above or on the red line satisfies $3x_1 + 5x_2 \geq 22$.

What this means: All these bundles cost at least as much as the old bundle $(4, 2)$ at the new prices $(3, 5)$.

If he chooses one of these bundles now, he **reveals** he can afford at least \$22, which means $(4, 2)$ is currently affordable but he's rejecting it.

4.3 The Purple DANGER ZONE (WARP Violation)

The crosshatched purple region is where **BOTH constraints hold**:

- Below the blue line (was affordable before)

- Above the red line (costs enough now)

Any bundle in this zone creates a logical contradiction:

1. **Past behavior:** He chose $(4, 2)$ over this bundle $\Rightarrow$ prefers $(4, 2)$
2. **Current behavior:** He chooses this bundle over $(4, 2) \Rightarrow$ prefers this bundle
3. **Contradiction!** Can't prefer both A to B and B to A

4.4 Example Bundles That Would Get Him Committed

- $(6, 1)$: Was affordable before (cost only \$9), but now costs \$23 > \$22 × WARP violation
- $(7, 1)$: Was affordable before (cost only \$10), but now costs \$26 > \$22 × WARP violation
- $(8, 0.5)$: Was affordable before (cost only \$9.5), but now costs \$26.50 > \$22 × WARP violation
- $(5, 2)$: Was NOT affordable before (cost \$11 > \$10) ✓ Safe - no violation
- $(3, 2)$: Was affordable (cost \$9) but only costs \$19 < \$22 now ✓ Safe - no violation

5 Why This Gets Him Committed

If the elderly man purchases any bundle in the danger zone, his family can argue:

> *"Your Honor, our father is not of sound mind. He previously rejected bundle X when it was affordable, proving he didn't want it. But now he's buying bundle X even though his original choice is still available and affordable. This is irrational, inconsistent behavior - clear evidence he cannot make sound financial decisions."*

The court would see this as a violation of basic economic rationality (WARP), potentially supporting a claim of mental incompetence.

5.1 The Clever Escape

To avoid being committed, he should:

- Stay BELOW the red line (buy cheaper bundles that don't signal a large budget), OR
- Stay ABOVE the blue line (buy bundles that weren't affordable before, so no revealed preference contradiction)
- **Definitely avoid the purple danger zone!**